DECLARATION

The internship report entitled **"FULL STACK DEVELOPMENT INTERNSHIP AT VYOMITE IT SERVICES",** submitted to the Department of Computer Science and Information Technology, Orchid International College, Gaushala, Kathmandu, Nepal under the supervision of **Mr. Naresh Yadav**, is an original work carried out during my internship period. This report has not been submitted earlier, in part or in full, to any university or institution, here or elsewhere, for the award of any degree.

Sincerely,

Gaurav Lamichhane (28914/078)

MENTOR RECOMMENDATION

This is to certify that **Mr. Gaurav Lamichhane** has successfully carried out the internship project entitled **"FULL STACK DEVELOPMENT INTERNSHIP AT VYOMITE IT SERVICES"** in partial fulfillment of the requirements for the Bachelor's Degree in Computer Science and Information Technology under Tribhuvan University, under my mentorship.

……

Mr. Subodh Shrestha

Co-Founder / Internship Mentor

Vyomite IT Services

Thulobharyang, Kathmandu, Nepal

ORCHID INTERNATIONAL College

SUPERVISOR RECOMMENDATION

This is to certify that **Mr. Gaurav Lamichhane** has carried out the internship project entitled **"FULL STACK DEVELOPMENT INTERNSHIP AT VYOMITE IT SERVICES"** in partial fulfillment of the requirements for the Bachelor's Degree in Computer Science and Information Technology under Tribhuvan University under my supervision.

To the best of my knowledge, this work has not been submitted to any other degree. The author has fulfilled all the requirements laid down by Orchid International College, Department of Computer Science and Information Technology, Gaushala, Kathmandu, Nepal.

……..

Mr. Naresh Yadav

Supervisor / Lecturer

Department of Computer Science and Information Technology

Orchid International College

Gaushala, Kathmandu, Nepal

ORCHID INTERNATIONAL College

LETTER OF APPROVAL

This is to certify that Gaurav Lamichhane (TU Exam Roll No. 28914/078) has successfully completed his internship at Vyomite IT Services as a Full-Stack Developer Intern. This internship report has been reviewed and approved and is hereby recommended to the Department of Computer Science and Information Technology for acceptance. The report has been prepared by the students for academic purposes and is submitted as partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Information Technology (BSc. CSIT).

…… **Mr. Dhiraj Kumar Jha** HOD / Coordinator, Department of IT, Orchid International College, Gaushala, Kathmandu, Nepal	…… **Mr. Naresh Yadav** Supervisor / Lecturer, Department of IT, Orchid International College, Gaushala, Kathmandu, Nepal
…… **Mr. Janak K. Lal** Fulltime Faculty, Department of IT, Orchid International College, Gaushala, Kathmandu, Nepal	…… **External Examiner** Central Department of Computer Science and Information Technology, Tribhuvan University, Kritipur, Nepal

COMPLETION LETTER

VYOMITE

Regd No: 012345/090/081 Pan No: 129893707

Date: 24 Feb, 2026

To Whom It May Concern,

Dear Sir/Madam

This is to certify that Mr. Gaurav Lamichhane has successfully completed his internship as a Full Stack Developer. During this period, he gained practical experience in both frontend and backend development, contributing to the overall design and functionality of web applications.

Throughout his internship, he was involved in developing user interfaces, integrating APIs, managing databases, and implementing server-side logic. He worked with modern development tools and technologies to ensure application performance, security, and scalability. His ability to understand system architecture and handle end-to-end development tasks demonstrated strong technical competence and analytical thinking.

Mr. Gaurav showed dedication, professionalism, and a continuous eagerness to learn. We are confident that the knowledge and hands-on experience he acquired will greatly support his future career as a Full Stack Developer.

If you require any additional information or documentation regarding this internship, please feel free to contact us.

Warm Regards,

Vyomite IT Services Thulobharyang, Kathmandu, Nepal

Subodh Shrestha

Project Lead

+977-9813453246 vyomite.com.np info@vyomite.com.np Thulobharyang, KTM

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to all those who have supported and guided me throughout the completion of this internship project. This work would not have been possible without the encouragement, guidance, and support of many individuals. I would like to express my sincere thanks to **Mr. Dhiraj Kumar Jha**, Program Coordinator of the Department of Computer Science and Information Technology, Orchid International College, for providing the necessary facilities and support throughout the internship period.

I am deeply thankful to **Mr. Subodh Shrestha**, my internship supervisor, for his invaluable guidance, advice, and encouragement throughout the course of this project. His insights, critical feedback, and continuous support greatly helped me improve the quality of my work and stay focused.

I would also like to extend my appreciation to the staff at Vyomite IT Services, especially the technical and administrative team, for providing me with the necessary facilities and assistance during my internship. Special thanks to **Mr. Subodh Shrestha**, Co-Founder of Vyomite IT Services, for providing me with the opportunity to apply theoretical knowledge from my bachelor's studies into practical implementation.

Finally, I am immensely grateful to my family and friends for their constant encouragement, patience, and support, which gave me the motivation to persevere and complete this internship successfully.

Sincerely,

Gaurav Lamichhane (28914/078)

ABSTRACT

In the rapidly evolving digital landscape, establishing a strong online presence is essential for organizations seeking to enhance visibility and engagement. This internship project focuses on the full-stack development of the Vyomite website, designed to showcase the company's services and strengthen its digital presence. The frontend was developed using Next.js, enabling a modern, responsive, and high-performance user interface that ensures accessibility across multiple devices and an intuitive user experience. The backend was implemented using Django, providing a robust and scalable environment for application logic and RESTful API development, while PostgreSQL was used as the database management system to ensure secure, reliable, and efficient data storage and retrieval. The integration of these technologies demonstrates the practical application of full-stack web development principles and highlights the importance of scalable architecture, efficient API design, and responsive user interfaces in building modern web applications for business environments.

Keywords: *Full-Stack Development, Next.js, Django, PostgreSQL, RESTful API, Web Application Development, Responsive Design.*

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LIST OF ABBREVIATIONS

API	Application Programming Interface
CLI	Command Line Interface
CORS	Cross Origin Resource Sharing
CSP	Content Security Policy
CRUD	Create, Read, Update, Delete
CSR	Client-Side Rendering
DBMS	Database Management System
DOM	Document Object Model
DRF	Django REST Framework
ERD	Entity Relationship Diagram
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IP	Internet Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token
MVC	Model-View-Controller
MVT	Model-View-Template
OAuth	Open Authorization
ORM	Object Relational Mapping
SEO	Search Engine Optimization
SQL	Structured Query Language
SSG	Static Site Generation

SSR	Server-Side Rendering
UI	User Interface
UX	User Experience

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CHAPTER 1 INTRODUCTION

1.1 Introduction

In today's digital era, a strong online presence is crucial for IT companies to showcase their services, attract clients, and maintain competitiveness. Vyomite IT Services, like many growing organizations, requires a modern, responsive, and interactive website to represent its capabilities, provide information to potential clients, and facilitate business engagement. Prior to this project, the company lacked a comprehensive digital platform, which limited its visibility and ability to efficiently share its services with clients.

The objective of this internship project was to design and develop full-stack web applications addressing both external business needs and internal operational efficiency. The first part of the project involved developing a corporate website for Vyomite IT Services. This included creating a modern and responsive frontend using Next.js, a robust backend using Django, and a secure and scalable PostgreSQL database to manage dynamic data. The system was designed to provide a seamless user experience while enabling easy management of company information and service offerings.

In addition to the corporate website, a full-stack Task Management Platform was developed to streamline task organization and improve productivity. This platform was built using React.js and Next.js for the frontend, and Django with Django REST Framework (DRF) for the backend. It allows users to register, authenticate, create, update, assign, and manage tasks efficiently. The system includes features such as task prioritization, status tracking, search and filtering, audit logs, and real-time updates, supported by secure JWT-based authentication and a PostgreSQL database for persistent storage.

These applications not only serve as a digital gateway for potential clients but also provide efficient tools for internal task management and communication. By integrating modern technologies, the project demonstrates the practical application of full-stack development principles, highlighting the importance of responsive design, scalable architecture, secure authentication, and efficient data management in modern web application development.

The successful implementation of both the Vyomite website and the Task Management Platform enhances the organization's online visibility, strengthens its professional credibility, and improves internal workflow efficiency. Furthermore, it provides a strong foundation for future digital growth, scalability, and effective client engagement.

1.2 Problem Statement

Limited or manual digital management of company information can lead to reduced visibility, inefficient communication with clients, and missed business opportunities. Vyomite IT Services did not have a centralized, interactive platform to showcase its services, update projects, or manage client inquiries effectively.

This made it difficult to maintain a professional online presence and engage potential clients efficiently. There was a pressing need for a user-friendly, feature-rich, and scalable website that could streamline the presentation of company services, enable seamless client interaction, and simplify content management.

Integrating a full-stack website solution using Next.js for the frontend, Django for the backend, and PostgreSQL for database management addresses these challenges by providing a secure, responsive, and interactive digital platform. This website allows the company to efficiently manage and update content, showcase services, communicate with clients, and improve overall business operations, ultimately enhancing the company's online presence and credibility.

1.3 Objectives

The objectives of the internship can be summarized as follows:

- To develop a full-stack website for Vyomite IT Services that allows efficient management and presentation of company information, including services, projects, announcements, and client interactions.
- To design and implement robust backend functionalities using Django, providing secure data management, API endpoints for dynamic content, and seamless integration with the frontend.
- To build a modern, responsive frontend using Next.js, ensuring a user-friendly interface, smooth navigation, and accessibility across multiple devices.
- To integrate PostgreSQL database solutions to support secure, efficient storage and retrieval of company data.
- To demonstrate the practical application of full-stack development principles, combining frontend and backend technologies to create a scalable, interactive, and professional digital platform.

1.4 Scope and Limitation

The Vyomite website project aimed to develop a comprehensive digital platform for Vyomite IT Services, enabling the management and presentation of company information, services, projects, and client interactions. The project focused on improving the company's online presence, providing a professional and interactive environment for clients and stakeholders. Specifically, the scope included the development of a responsive frontend using Next.js, a robust backend using Django, and a secure PostgreSQL database to handle dynamic content. Additionally, the project involved creating backend APIs to ensure seamless integration between frontend and backend systems and efficient data management.

The limitations of the project include potential technical challenges in ensuring smooth communication between frontend and backend components, optimizing performance across different devices, and maintaining secure and reliable data storage. The success of the website depends on its usability, responsiveness, and the company's ability to update content efficiently. Ensuring data security, protecting sensitive company information, and accommodating future feature expansions were key considerations during development.

1.5 Report Organization

This internship report consists of four chapters. The report has been organized in the order given below:

Chapter 1: Introduction

Introduces the internship report, covering the project background, problem statement, objectives, scope, limitations, and report organization.

Chapter 2: Organization Details and Literature Review

Presents the host organization, Vyomite IT Services, including its hierarchy, working domains, and internship department.

Chapter 3: Internship Activities

Details the internship work, including roles, responsibilities, weekly log, project implementation, and tasks completed.

Chapter 4: Conclusion and Learning Outcomes

Summarizes the internship, highlighting achievements, challenges, learning outcomes, and reflective conclusions.

References and Appendices have been listed at the end of the report under different sections.

CHAPTER 2 ORGANIZATION DETAILS AND LITERATURE REVIEW

2.1 Introduction to Organization

Vyomite IT Services is a dynamic and innovative IT company specializing in delivering high-performance software solutions tailored to the unique needs of its clients. The dedicated team of experienced professionals is committed to providing top-quality services while adhering to strict timelines and cost-effective approaches. Over the years, Vyomite has served a diverse range of clients, from small enterprises to larger organizations, offering expertise across web development, mobile application development, UI/UX design, digital marketing, branding, and other IT services.

At the core of Vyomite's mission lies a commitment to excellence, client satisfaction, and delivering measurable results. The company emphasizes a client-centric approach, leveraging skilled professionals, industry-specific solutions, and modern technologies to provide reliable and scalable IT solutions. This ensures enduring business value for all clients while fostering innovation, efficiency, and professional credibility in every project undertaken.

Table 2.1 Overview of company profile

Organization Name	Vyomite IT Services
Address	Thulobharyang, Kathmandu
Contact	+977-9813453246
Year of establishment	2023
Email	info@vyomite.com.np
Website	https://vyomite.com.np/

Table 2.2 Overview of Internship

Start Date	1st Dec 2025
End Date	1st March 2026
Total Duration	3 Months
Working Days	6 days a week
Intern Position	Full-Stack Intern
Mentor	Mr. Subodh Shrestha
Office Hour	10:00 AM to 5:00 PM

2.2 Organization Hierarchy

The hierarchical structure of an organization plays a vital role in ensuring its smooth and effective operation. Such structures define how work is divided, how accountability is distributed, and how communication travels across different levels acting as the foundation that keeps all parts of the organization aligned and moving toward shared goals.

Vyomite IT Services operates within a well-defined functional hierarchy designed to support its vision, objectives, and strategic goals. As illustrated in Figure 2.1, the organizational chart begins with the CEO at the top, who provides overall direction and leadership. Reporting directly to the CEO is the CTO, who oversees all technical operations and strategy. Under the CTO, the Tech Lead serves as the bridge between leadership and the development teams, coordinating technical execution across disciplines.

The Tech Lead directly manages three specialized senior roles the Senior Frontend Developer, Senior UI/UX Designer, and Senior Backend Developer each responsible for their respective domain of expertise. Supporting these senior roles is the Intern, who contributes to the team's work while gaining hands-on experience within the organization.

This structure provides a clear roadmap for role definition, reporting lines, and decision-making at Vyomite IT Services, ensuring that every team member understands their responsibilities and how they contribute to the organization's broader mission.

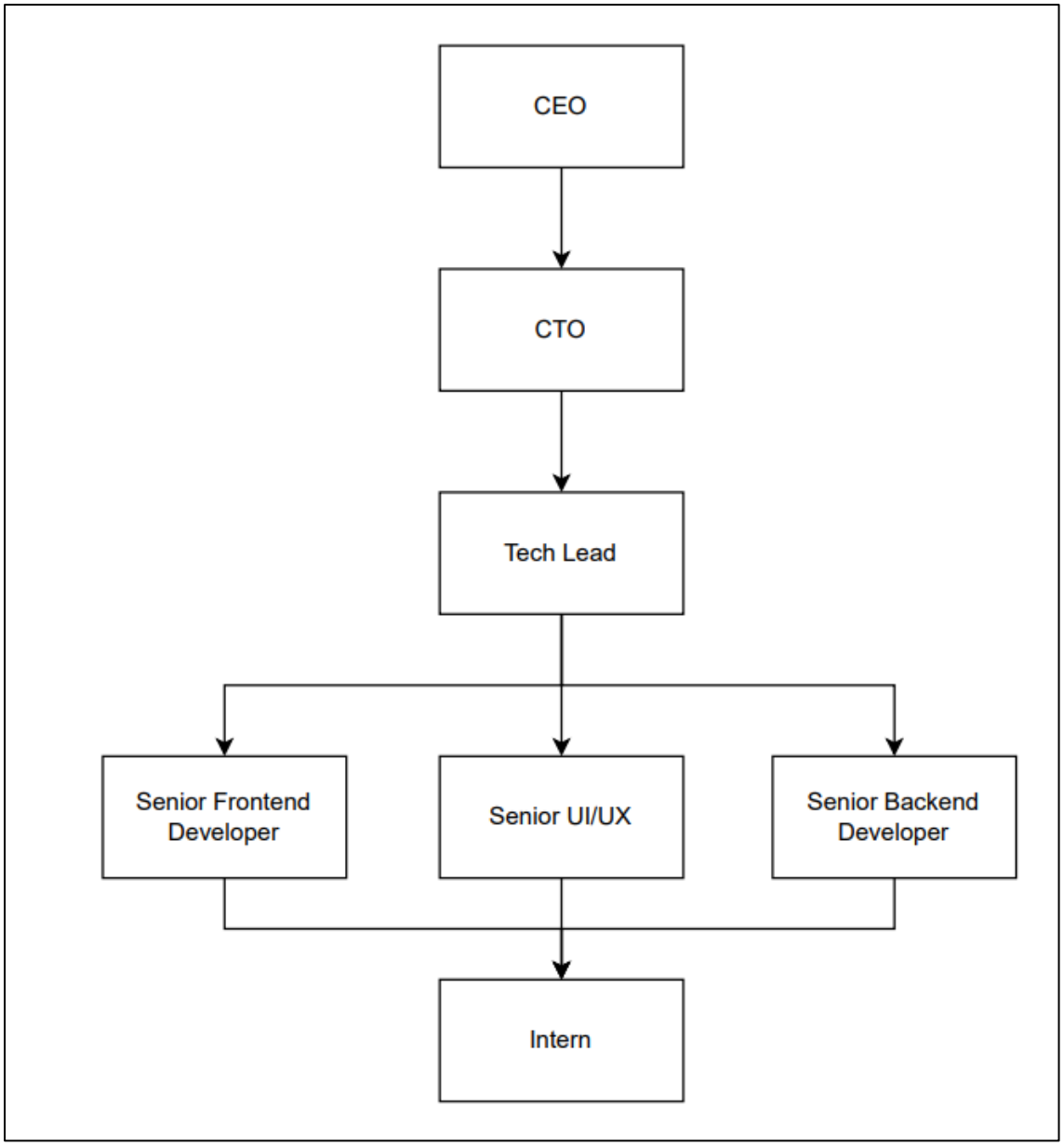

Figure 2.1 Organizational Hierarchy

2.3 Working Domains of Organization

Vyomite IT Services operates across multiple IT and software development domains, providing comprehensive solutions to meet the diverse needs of its clients. The key working domains include:

2.3.1 Web Development

Vyomite specializes in both static and dynamic website development. The team leverages modern frameworks and technologies, including Next.js, Django, and responsive web design principles, to create scalable, high-performance, and user-friendly websites tailored to client requirements.

2.3.2 Mobile Application Development

The organization develops mobile applications across Android and iOS platforms, using technologies such as Flutter, React Native, and native development frameworks. The focus is on building interactive, efficient, and secure mobile solutions that enhance user experience.

2.3.3 UI/UX Design

Vyomite provides user-centric UI/UX design services, ensuring visually appealing and intuitive interfaces. Designs are crafted to maintain brand consistency, enhance usability, and deliver engaging digital experiences across web and mobile platforms.

2.3.4 Digital Marketing

The company delivers tailored digital marketing services, including SEO, social media handling, content marketing, and branding, to enhance online visibility, drive customer engagement, and maximize conversion rates.

2.3.5 Hosting and IT Support

Vyomite offers reliable hosting solutions and 24/7 technical support, ensuring secure and uninterrupted operation of client websites and applications.

2.4 Description of Internship Department/Unit

The Web Development Unit at Vyomite IT Services serves as the central hub for technology-driven projects, specializing in the development and maintenance of web applications. As a Full-Stack Development Intern, I was integrated into a team of skilled developers working on the Vyomite website project.

During the internship, hands-on experience was gained in frontend development using Next.js, backend development with Django, and database management using PostgreSQL. The work included designing APIs, implementing business logic, optimizing performance, and ensuring secure data handling. Collaboration with designers, project managers, and

other developers was an integral part of the internship, contributing to live projects and applying problem-solving skills to real-world challenges. This experience provided a strong foundation for professional growth and practical exposure to full-stack development in a real-world IT company environment.

2.5 Literature Review

Modern web applications rely heavily on robust frameworks and efficient development practices to ensure scalability, maintainability, and performance. Full stack development involves integrating frontend technologies, backend frameworks, and databases to build complete web applications. Various studies and technical documentations highlight the importance of modern frameworks and structured development approaches in building efficient software systems.

Next.js is a popular React-based framework designed to simplify frontend development by providing features such as server-side rendering, static site generation, and built-in routing. According to the Next.js documentation, the framework improves web application performance and developer productivity by offering optimized rendering techniques and automatic code splitting. These features help developers build fast, SEO-friendly, and scalable web applications while maintaining a modular component-based architecture [1].

On the backend side, Django is a widely used high-level Python web framework that promotes rapid development and clean, pragmatic design. The Django documentation explains that the framework follows the Model-View-Template (MVT) architectural pattern, which separates application logic, user interface, and database operations. Django provides built-in features such as authentication, security protection, and ORM (Object Relational Mapping), allowing developers to build secure and maintainable web applications efficiently [2].

To facilitate communication between frontend and backend systems, modern web applications commonly use RESTful APIs. Django REST Framework (DRF) is an extension of Django that simplifies the development of REST APIs by providing tools for serialization, authentication, request handling, and API view management. According to the Django REST Framework documentation, DRF enables developers to easily create scalable APIs that allow different systems or client applications to interact with backend services efficiently [3].

In addition to selecting appropriate technologies, maintaining proper coding standards and development practices is essential for building reliable software systems. Coding standards help ensure consistency, readability, and maintainability of code within development teams. DevCom highlights that following coding standards and best practices can significantly reduce software errors, improve collaboration among developers, and enhance the long-term sustainability of software projects. Practices such as modular programming, proper documentation, and consistent naming conventions contribute to more efficient software development processes [4].

Overall, these technologies and development practices form the foundation of modern full stack web development. Frameworks like Next.js and Django enable developers to build efficient frontend and backend systems, while tools like Django REST Framework facilitate seamless integration between them through APIs. Adhering to coding standards further ensures that applications remain maintainable, scalable, and reliable in real-world software development environments.

CHAPTER 3 INTERNSHIP ACTIVITIES

3.1 Roles and Responsibilities

During the internship at Vyomite IT Services, the author served as a Full-Stack Development Intern, contributing significantly to the web development team. The role involved designing, developing, and maintaining the Vyomite website, integrating frontend and backend components, and ensuring smooth functionality across the platform.

The responsibilities of the Full-Stack Development Intern included:

- Collaborating with the development team to design and implement backend solutions using Django and Python.
- Developing a responsive frontend using Next.js, ensuring seamless integration with backend services.
- Managing and interacting with the PostgreSQL database, including designing schemas, querying data, and ensuring data integrity.
- Creating and maintaining APIs to facilitate smooth communication between the frontend and backend.
- Collaborating with UI/UX designers and frontend developers to ensure consistent and user-friendly interfaces.

Participating in problem-solving sessions to identify and address technical issues, optimizing application performance and security.

3.2 Weekly Log

The log below was maintained throughout the internship period from 1st Dec 2025 to 1st March 2026. The log contains details about tasks performed each week during my intern period.

Table 3.1 Internship Weekly Log

Week	Activities Performed
1st Week	Became familiar with the organization's policies, work culture, and development workflow. Configured the required development tools and environment for the

	internship role. Established the initial setup necessary for starting the project development.
2nd Week	Conducted research on coding standards and best practices to ensure maintainable code. Explored Next.js framework, including its file structure and project organization. Reviewed Git and GitHub workflows and revised React concepts such as hooks, routing, props, and context.
3rd Week	Set up the project environment for frontend and backend development. Studied and analyzed the Figma design to align implementation with the UI/UX requirements. Created reusable components such as the navbar, footer, and call-to-action sections to ensure consistency across the website.
4th Week	Applied Figma design guidelines more accurately to the user interface for better visual consistency. Gained a deeper understanding of component-based architecture. Translated Figma designs into fully responsive layouts for multiple screen sizes.
5th Week	Integrated Leaflet maps to enhance location-based features on the website. Implemented service filtering functionality on the service page for better user experience. Developed dynamic routing for service-specific pages to enable smooth navigation and content display.

6th Week	Added reCAPTCHA to selected pages to enhance security and prevent spam. Implemented skeleton loaders for improved user experience during data loading. Replaced standard image tags with Next.js Image component for optimized performance. Integrated React forms and conducted research on microanimations to enhance UI interactivity.
7th Week	Gained understanding of the password reset flow via email for user account management. Started and structured the backend project for scalable development. Developed database models and successfully integrated the frontend with backend services to ensure smooth communication.
8th Week	Implemented IP-based rate limiting in the backend to enhance security and prevent abuse. Conducted research on OpenID and OAuth authentication protocols for secure user login. Developed and integrated audit logging in the backend to track critical system activities.
9th Week	Gained practical understanding of Docker concepts and containerization for application deployment. Practiced creating and managing containers using Docker CLI commands. Fixed the hamburger menu issue and resolved other UI-related bugs to improve user experience.
10th Week	Enhanced backend security mechanisms to safeguard application data. Implemented

	Google reCAPTCHA v2 across the application to prevent automated attacks. Fixed localStorage errors on the frontend to ensure smooth client-side functionality.
11th Week	Integrated Cloudinary to manage image-related fields efficiently. Prepared comprehensive API documentation for backend services. Resolved Content Security Policy (CSP) related errors to ensure application compliance and security.
12th Week	Fixed inconsistencies in browser rendering, particularly on Microsoft Edge. Improved overall website performance for a smoother user experience. Optimized page loading times to enhance efficiency and responsiveness.
13th Week	Successfully completed and delivered the full project. Received positive feedback on both UI/UX design and technical implementation.

3.3 Description of the Project Involved During the Internship

This section offers a comprehensive overview of the projects, presenting a holistic view of the practical applications of acquired skills and knowledge throughout the internship tenure.

3.3.1 Vyomite Website (Company Website)

As part of the internship training, a modern IT Company Website was developed using a full-stack architecture consisting of Next.js for the frontend and Django with Django REST Framework (DRF) for the backend. The objective of this project was to gain practical experience in designing scalable web applications, implementing RESTful APIs, and integrating frontend interfaces with backend services.

The frontend was developed using Next.js, a React-based framework that enables high-performance web applications through features such as server-side rendering, static site generation, and optimized routing. The website provides sections for company services, blog posts, career opportunities, team information, and contact forms through a responsive and user-friendly interface. The application utilized the Next.js App Router for dynamic page navigation and implemented a component-based architecture to ensure reusable and maintainable UI elements. Additional features such as automatic font optimization using next/font, TypeScript for type safety, and PostCSS with modern CSS techniques were incorporated to improve performance, reliability, and styling consistency.

The backend system was developed using Django and Django REST Framework, enabling the creation of scalable and well-structured RESTful API services. The backend was organized into modular Django applications including blog, careers, contact, security, and audit, each responsible for managing specific functionality. Database models were designed to represent relationships between entities such as blog posts, job postings, and user interactions, and Django ORM was used to perform secure and efficient database operations.

Security measures were implemented throughout the system, including authentication, authorization, input validation, and CORS configuration to protect API endpoints. Sensitive configurations were managed using environment variables to support secure deployment across different environments. The audit module was implemented to maintain logs of important system activities, improving transparency and debugging capabilities.

All API endpoints were designed following RESTful principles, implementing proper HTTP methods, status codes, validation, and error handling to ensure reliable communication between the frontend and backend components.

Overall, the development of this project provided practical experience in full-stack web development, including modern frontend frameworks, REST API architecture, database design, authentication and authorization, and scalable application structure. The project demonstrated how modern technologies can be integrated to build efficient, secure, and maintainable web applications.

3.3.2 Task Management Platform

As part of the internship training, a Task Management Platform was developed using Django REST Framework (DRF) for the backend and React.js for the frontend. The

objective of this project was to gain practical experience in building full-stack web applications and understanding how frontend interfaces communicate with backend services through RESTful APIs.

The system allows users to register, authenticate, and manage tasks efficiently. Users can create, update, delete, and organize tasks with additional features such as task assignments, status tracking, and audit logging. The frontend was implemented using React.js with Tailwind CSS, providing a responsive and user-friendly interface.

The backend was built using Django 5.2 and Django REST Framework, enabling the development of scalable API endpoints. PostgreSQL was used as the database to ensure reliable data storage and integrity. Secure authentication and role-based permissions were implemented to control user access and actions.

The backend architecture follows Django's Model–View–Serializer (MVS) pattern, ensuring proper separation of concerns. On the frontend, React's useState and useEffect hooks were used for state management and asynchronous data fetching, allowing the interface to update dynamically as users interact with the system.

Additionally, API documentation was automatically generated using drf-spectacular, providing interactive documentation through Swagger UI and ReDoc. This allowed developers to easily explore API endpoints, request parameters, and response formats.

Overall, the project provided practical experience in full-stack development, including RESTful API design, database modeling, authentication, frontend–backend communication, and responsive UI design.

3.3.3 Tools & Technologies Used

Languages & Frameworks:

- HTML5, CSS3, JavaScript
- Python, Django, Django REST Framework
- Next.js
- React

Database:

- PostgreSQL
- SQLite

Development Tools:

- VS Code
- Git / GitHub
- Postman
- OpenAPI Swagger Documentation

3.3.4 Description of Major Technologies

3.3.4.i React.js

React.js is a JavaScript library used for building interactive and component-based user interfaces. It allows developers to create reusable UI components and efficiently update the interface using a virtual DOM. In this project, React was used to develop dynamic frontend components and manage user interactions.

3.3.4.ii Next.js

Next.js is a React-based framework used for building modern web applications with features such as server-side rendering, static site generation, and optimized routing. It was used to build the frontend of the IT Company Website, improving performance and SEO while providing seamless page navigation.

3.3.4.iii Django

Django is a high-level Python web framework that enables rapid development of secure and scalable web applications. It provides built-in features such as authentication, ORM, and an administrative interface. In this project, Django was used to build the backend system and manage business logic.

3.3.4.iv Django REST Framework (DRF)

Django REST Framework is a powerful toolkit used to build RESTful APIs with Django. It provides serializers, authentication mechanisms, and API views to simplify backend API development. DRF was used to create structured API endpoints for communication between the frontend and backend.

3.3.4.v PostgreSQL

PostgreSQL is an open-source relational database management system known for its reliability and performance. It was used to store and manage application data such as blog posts, career information, and user interactions.

3.3.4.vi Git & GitHub

Git is a version control system used to track code changes during development. GitHub was used as the remote repository for collaboration, code management, and version control throughout the project.

3.3.4.vii Postman

Postman is an API testing tool used to send HTTP requests and verify API responses. It was used to test backend endpoints during the development of the Django REST APIs.

3.3.4.viii Swagger (OpenAPI Documentation)

Swagger is an API documentation tool that automatically generates interactive documentation for REST APIs. In this project, Swagger was used to visualize and test API endpoints created using Django REST Framework.

3.4 Tasks / Activities Performed

In this section, we delve into the technical intricacies of the activities carried out throughout the internship. It offers detailed insight into the specific tasks, projects, and skills that were actively engaged during the internship period.

3.4.1 Learning and Practicing Full-Stack Development Fundamentals

As part of the internship training, fundamental concepts of full-stack web development were studied and practiced. This included learning about frontend component architecture, backend API development, database integration, and client–server communication in modern web applications.

Hands-on exercises involved building React and Next.js components, managing state using useState and useEffect, and fetching data from backend APIs. At the backend, Django and Django REST Framework (DRF) were used to create API endpoints and manage database models. These activities helped develop a practical understanding of how frontend and backend systems work together to build scalable web applications.

3.4.2 Development of Tasks Management Platform (Full-Stack Application)

During the internship, a full-stack Task Management Platform was developed using React.js and Next.js for the frontend and Django with Django REST Framework (DRF) for the backend. The objective was to build a complete task management system where users

could register, authenticate, create, update, assign, and organize tasks, with persistent storage and real-time updates.

The frontend was implemented using React.js with modern hooks such as useState and useEffect for state and side-effect management. It featured dynamic task creation and editing, real-time status updates, task filtering and searching, and a responsive interface styled with Tailwind CSS. Reusable components and custom hooks were used to promote modularity and maintainability.

The backend was developed using Django 5.2 and DRF, providing RESTful API endpoints for all CRUD operations on tasks and user management. JWT token-based authentication secured the system, while database models represented users, tasks, and audit logs. The backend also implemented permission-based access control, input validation, and comprehensive error handling to ensure reliability and security.

Key features of the platform included:

- **User Authentication:** Registration, login, and secure password handling with JWT tokens.
- **Task Management:** CRUD operations, task assignment, status tracking, priority levels, and due dates.
- **Advanced Features:** Audit logs for all task modifications, search and filter functionality, task analytics, and real-time notifications.

The development of this platform provided practical experience in full-stack application architecture, frontend-backend communication via REST APIs, database design, authentication and authorization, responsive UI design, and asynchronous programming in JavaScript. It strengthened skills in building scalable, secure, and user-friendly web applications.

3.4.2.i Snippet from React Task Dashboard State Management

```
function TaskList() {
  const [tasks, setTasks] = useState([]);
  const [filter, setFilter] = useState("all");
  const [searchTerm, setSearchTerm] = useState("");
  const [loading, setLoading] = useState(false);
```

```
  const [error, setError] = useState("");
  useEffect(() => {
    fetchTasks();
  }, []);
  const fetchTasks = async () => {
    try {
      setLoading(true);
     const response=await axios.get('http://127.0.0.1:8000/api/tasks/',{
        headers: {
        'Authorization':`Bearer ${localStorage.getItem('access_token')}`
        }
      });
      setTasks(response.data);
    } catch (err) {
      setError("Failed to fetch tasks");
    } finally {
      setLoading(false);
    }
  };
}
```

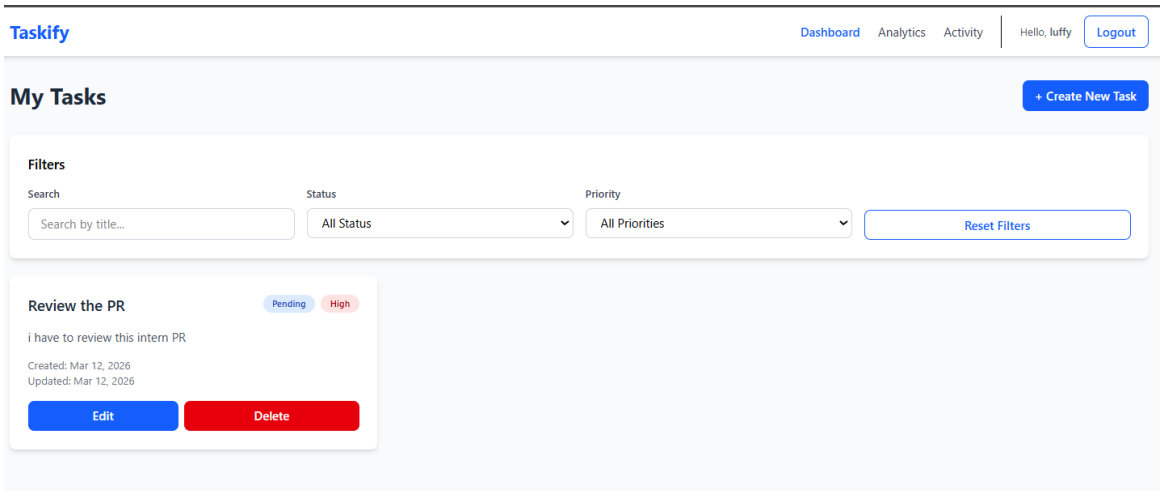

Figure 3.1 Task Management Platform Dashboard

3.4.2.ii Snippet from React Task Creation Form

```
function TaskForm() {
  const [formData, setFormData] = useState({
    title: "",
    description: "",
    priority: "medium",
    dueDate: "",
    assignedTo: ""
  });
  const [error, setError] = useState("");
  const [success, setSuccess] = useState("");
  const handleCreateTask = async (e) => {
    e.preventDefault();
    if (!formData.title || !formData.dueDate) return;
    try {
      awaitaxios.post('http://127.0.0.1:8000/api/tasks/', formData, {
        headers: {
        'Authorization':`Bearer ${localStorage.getItem('access_token')}`
        }
      });
      setSuccess("Task created successfully");
    } catch (err) {
      setError("Failed to create task");
    }
  };
}
```

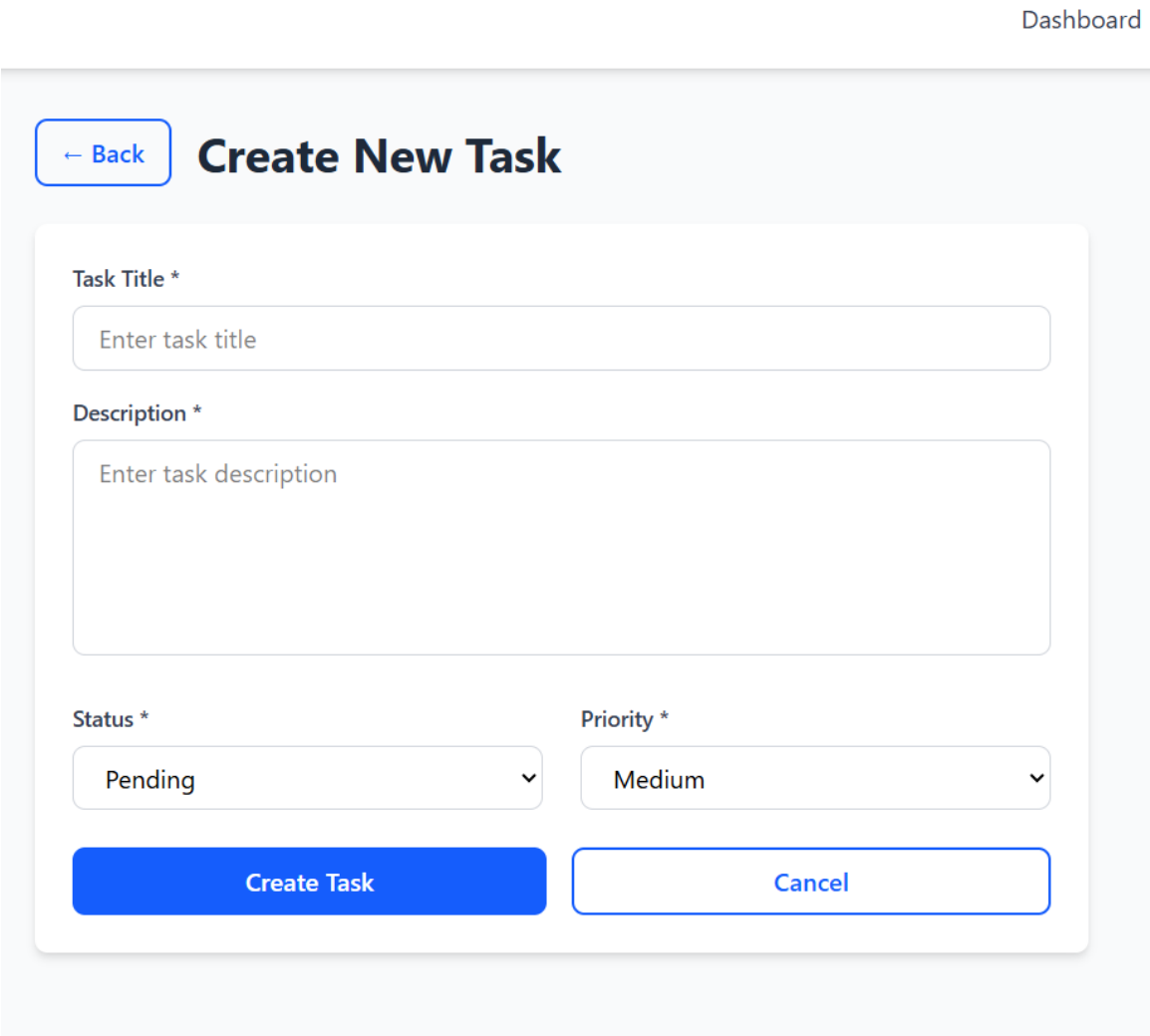

Figure 3.2 Task Creation Form

3.4.2.iii Snippet from Backend Task API Endpoint (Django REST Framework)

```
from rest_framework import viewsets
from rest_framework.permissions import IsAuthenticated
from .models import Task
from .serializers import TaskSerializer
class TaskViewSet(viewsets.ModelViewSet):
    serializer_class = TaskSerializer
    permission_classes = [IsAuthenticated]
    def get_queryset(self):
        return Task.objects.filter(assigned_to=self.request.user)
```

3.4.3 Development of IT Company Website (Vymite Website)

During the internship, a professional IT Company Website was developed using Next.js for the frontend and Django with Django REST Framework (DRF) for the backend. The objective was to build a fully dynamic website, where all content—including blogs, portfolio items, career listings, and contact submissions—is managed entirely from the backend, allowing real-time updates without redeploying the frontend.

The frontend was implemented using Next.js with React hooks for state and effect management. It features dynamic pages for blogs and careers, search and filter functionality, contact form submission with validation, and a fully responsive interface styled with modern CSS. Reusable components and modular architecture were applied for maintainability, scalability, and performance.

The backend was developed using Django and DRF, providing RESTful APIs for all content management operations. Database models represent blogs, portfolio items, career opportunities, and contact inquiries. Input validation, error handling, and audit logging were implemented to ensure reliable and consistent content delivery.

Key features of the website include:

- **Blog Management:** Create, edit, and delete posts; manage authors, categories, tags; publish/draft status; comment moderation.
- **Career Portal:** Job posting creation, application tracking, candidate information storage, and job status management.
- **Portfolio & Services**: Dynamic display of projects, services, and team information fetched from the backend.
- **Contact Management:** Form submission handling, email notifications, inquiry tracking, and response management.
- **Security & Reliability:** Input validation, error handling, and audit logging to maintain consistent data integrity.

The development of this website provided practical experience in full-stack development, including backend-driven dynamic content management, frontend-backend communication via REST APIs, responsive UI design, and professional web development practices. It strengthened skills in building scalable, maintainable, and fully dynamic web applications.

3.4.3.i Snippet from Frontend Blog Page with Dynamic Content

```
export default function BlogPage() {
  const [blogs, setBlogs] = useState([]);
  const [filteredBlogs, setFilteredBlogs] = useState([]);
  const [searchTerm, setSearchTerm] = useState('');
  const [selectedCategory, setSelectedCategory] = useState('all');
  useEffect(() => { fetchBlogs(); }, []);
  const fetchBlogs = async () => {
    try {
      const response = await axios.get('https://backendurl/api/blogs/');
      setBlogs(response.data);
      setFilteredBlogs(response.data);
    } catch (err) { console.error('Failed to load blogs'); }
  };
}
```

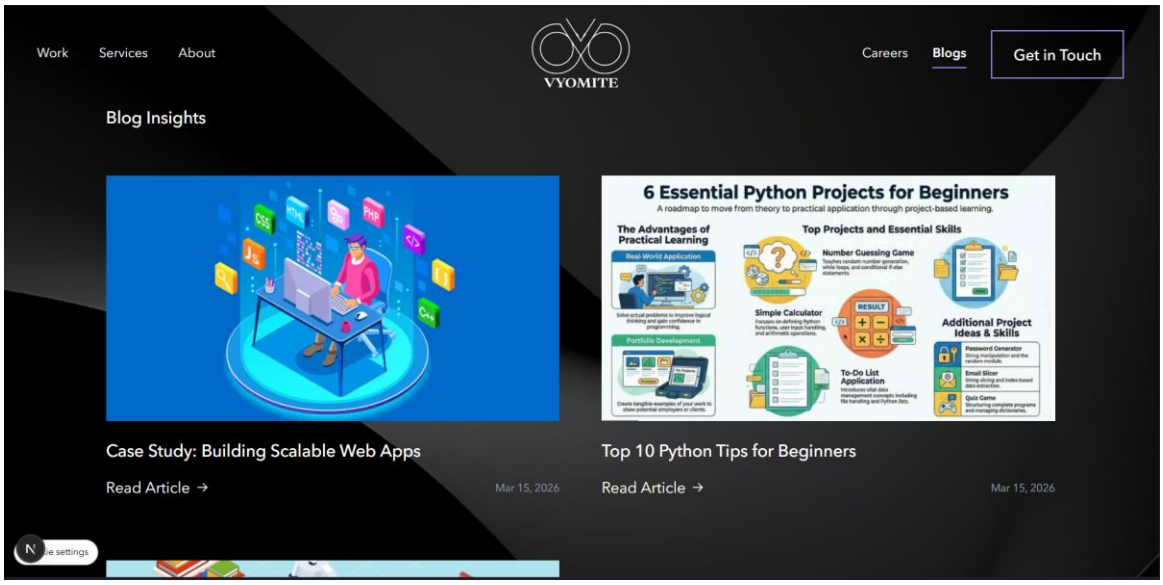

Figure 3.3 User interface of the Vyomite Blog Page

3.4.3.ii Snippet from Frontend - Careers Page

```
export default function CareersPage() {
  const [jobs, setJobs] = useState([]);
  const [searchTerm, setSearchTerm] = useState('');
  const [selectedDepartment, setSelectedDepartment] = useState('all');

  useEffect(() => { fetchJobs(); }, []);

  const fetchJobs = async () => {
    try {
      const response = await axios.get('https://backend-
      url/api/careers/jobs/');
      setJobs(response.data);
    } catch (err) { console.error('Failed to fetch jobs'); }
  };
}
```

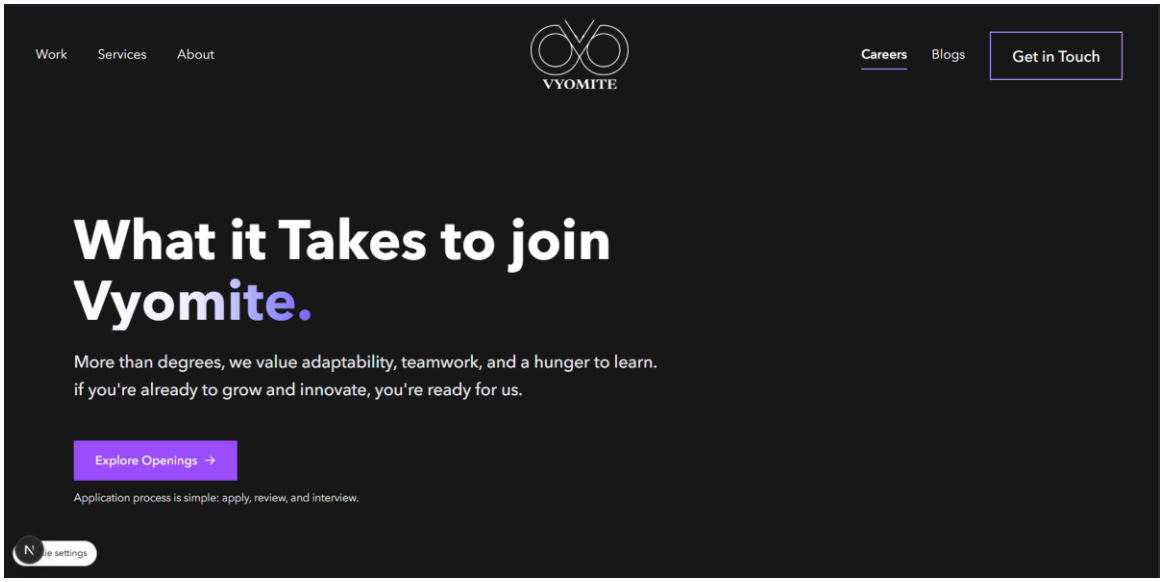

Figure 3.4 Careers Page of the Vyomite Website

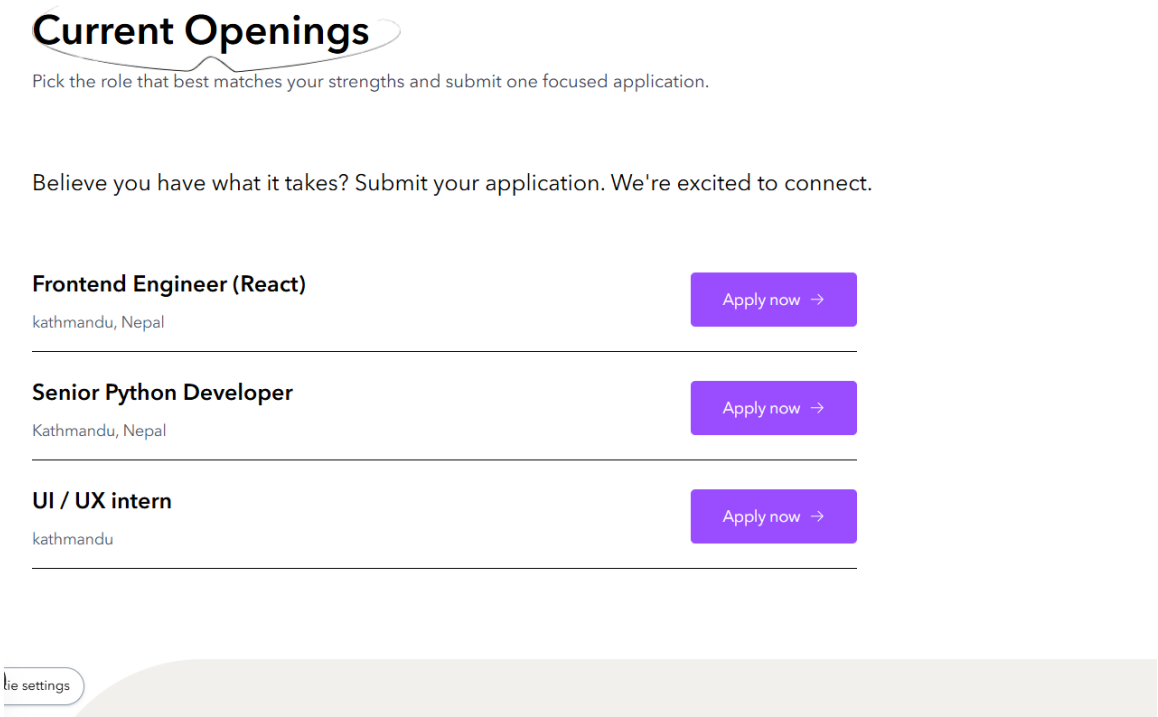

Figure 3.5 Job openings section on the Vyomite careers page

3.4.3.iii Snippet from Backend – Django API

```
from rest_framework import viewsets

from .models import Blog

from .serializers import BlogSerializer


class BlogViewSet(viewsets.ModelViewSet):

    queryset=Blog.objects.filter(status='published').order_by('-
    created_at')

    serializer_class = BlogSerializer
```

3.4.3.iv Snippet from Backend – Django Careers API

```
from rest_framework import viewsets

from .models import Job

from .serializers import JobSerializer

class JobViewSet(viewsets.ModelViewSet):

    queryset =Job.objects.filter(is_active=True).order_by('-created_at')

    serializer_class = JobSerializer
```

3.4.3.v Snippet from Frontend – Contact Form with Validation

```
export default function ContactForm() {
  const [formData, setFormData] = useState({name:'', email:'', message:''});
  const [errors, setErrors] = useState({});

  const handleSubmit = async (e) => {
    e.preventDefault();
    // Basic validation
    if (!formData.name || !formData.email || !formData.message) return;
    try { await axios.post('https://backend-url/api/contact/', formData); }
    catch (err) { console.error('Failed to submit form'); }
  };
}
```

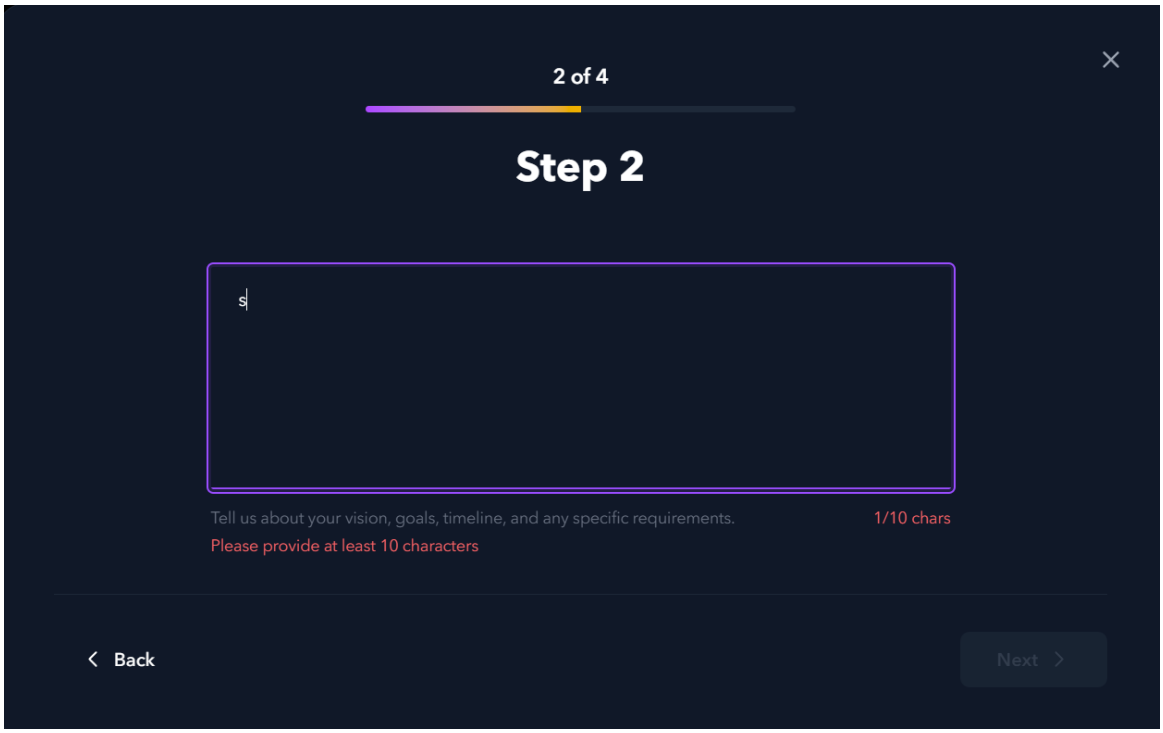

Figure 3.6 Client-side validation for the contact form in step 2

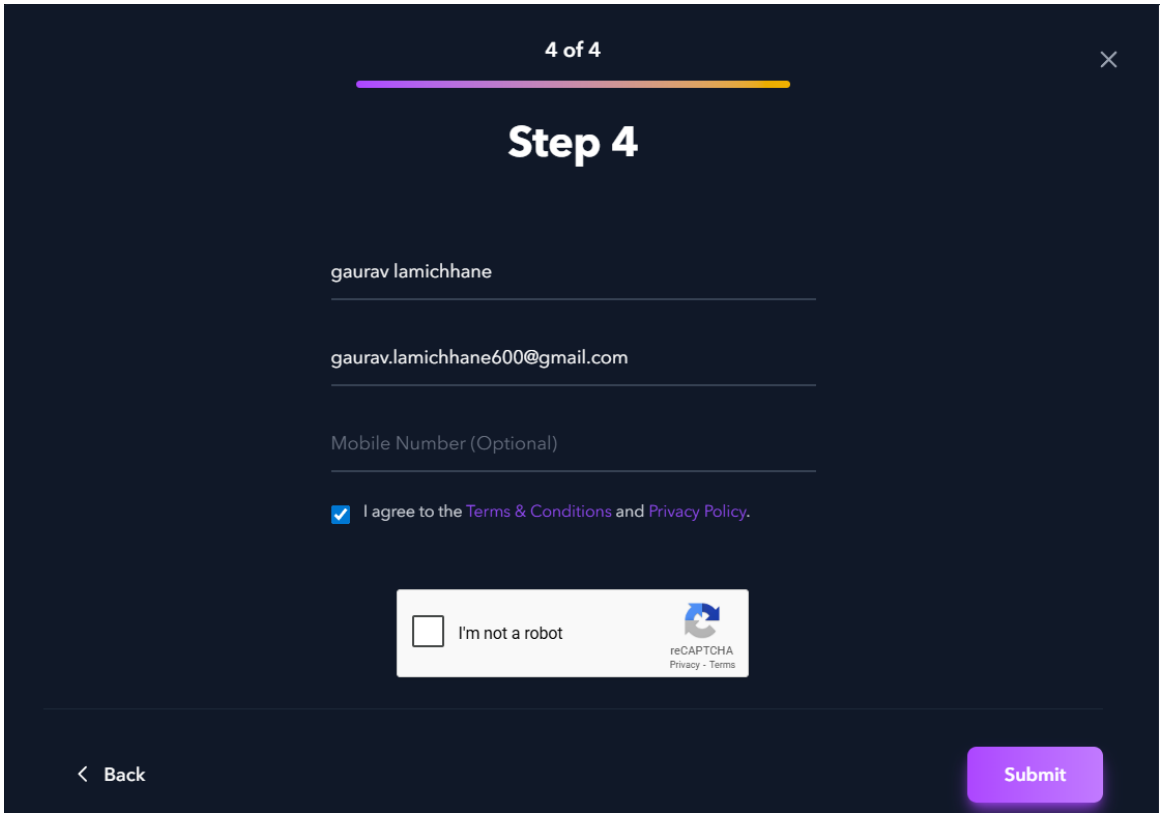

Figure 3.7 Client-side validation for the contact form in step 4

CHAPTER 4 CONCLUSION AND LEARNING OUTCOMES

4.1 Conclusion

The three-month internship at Vyomite IT Services was a valuable experience that contributed significantly to my professional and technical development. During the internship, I gained practical exposure to real-world software development practices while working on the development and enhancement of the Vyomite website.

Throughout the internship period, I improved my technical skills by working with technologies such as Next.js, Django, and PostgreSQL. The experience allowed me to understand the complete workflow of full-stack web development, including frontend design, backend implementation, API development, and database management. I also learned how different components of a web application interact to create a functional and efficient system.

In addition to technical knowledge, the internship helped me develop important professional skills such as teamwork, communication, and problem-solving. Working in a collaborative environment provided valuable insight into how software development teams coordinate tasks, manage projects, and deliver solutions effectively.

Although there were certain limitations, such as limited exposure to some advanced production-level systems, the internship provided a strong foundation in modern web development technologies and industry practices. Overall, the experience was highly beneficial and aligned well with my career goal of becoming a proficient full-stack software developer.

4.2 Learning Outcome

Through the internship at Vyomite IT Services, the following key learning outcomes were achieved:

4.2.1 Professional Level

- Developed the ability to work effectively in a team environment, collaborating with other developers and participating in project discussions and task coordination.
- Improved problem-solving, time management, and task prioritization skills while handling full-stack development tasks and resolving technical challenges.

- Gained insight into real-world software development workflows, including project planning, feature implementation, testing, and system optimization.
- Enhanced communication skills by coordinating with team members and understanding workplace responsibilities.

4.2.2 Technical Level

- Acquired practical experience in full-stack web development using Next.js for frontend and Django with DRF for backend development.
- Gained understanding of database management with PostgreSQL, including schema design, data storage, and retrieval.
- Developed skills in designing and integrating RESTful APIs to enable seamless communication between frontend and backend.
- Improved debugging and problem-solving skills by identifying and resolving issues in both frontend and backend components.
- Strengthened ability to create responsive, modular, and maintainable frontend components using React and Next.js.
- Learned how to manage dynamic content, implement search/filter functionality, and build fully backend-controlled web applications.

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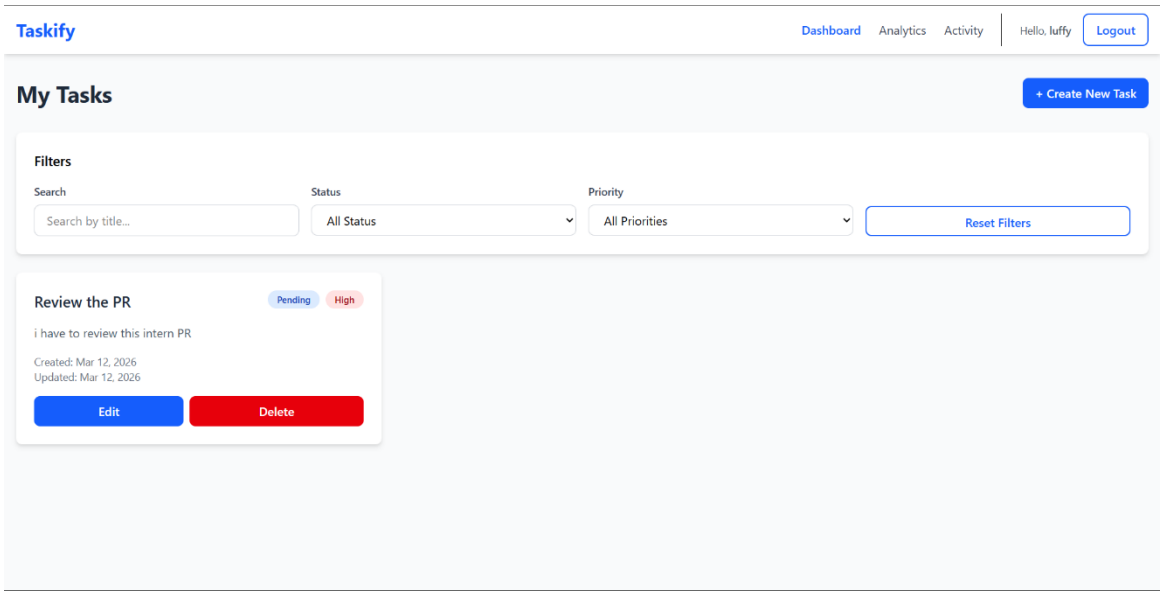
[3] Django REST Framework, “Django REST Framework Documentation.” [Online]. Available: https://www.django-rest-framework.org/.

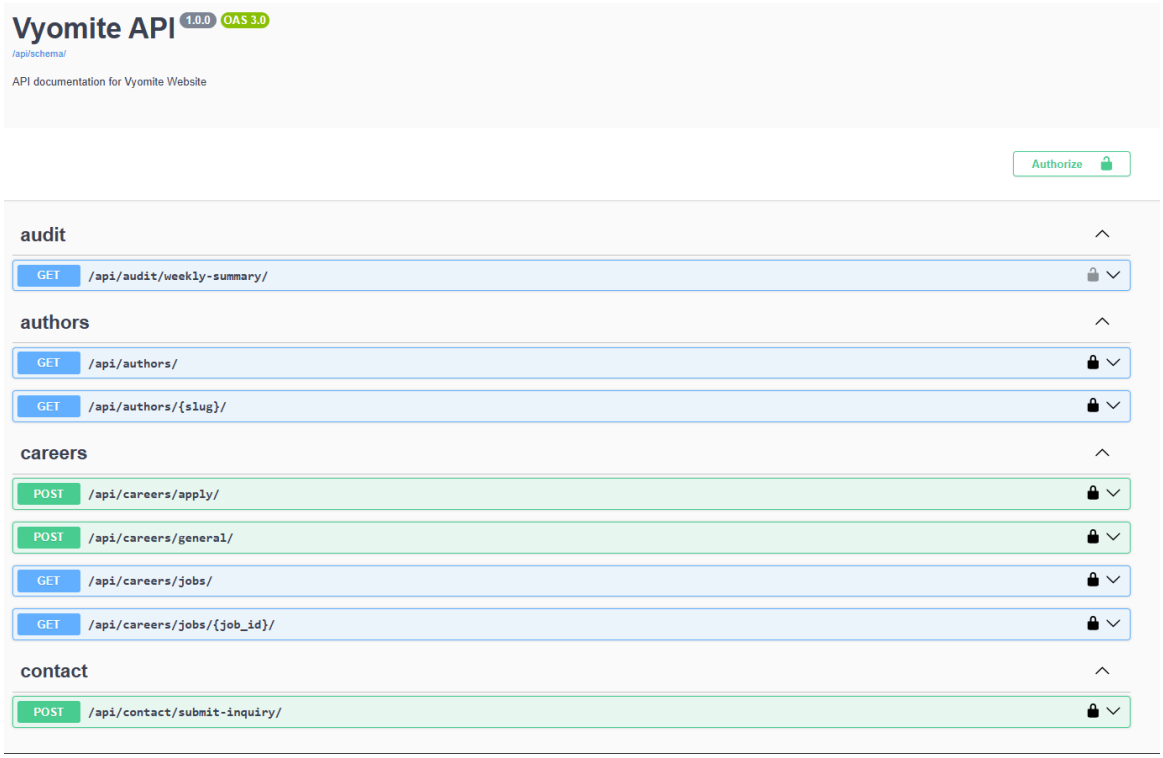
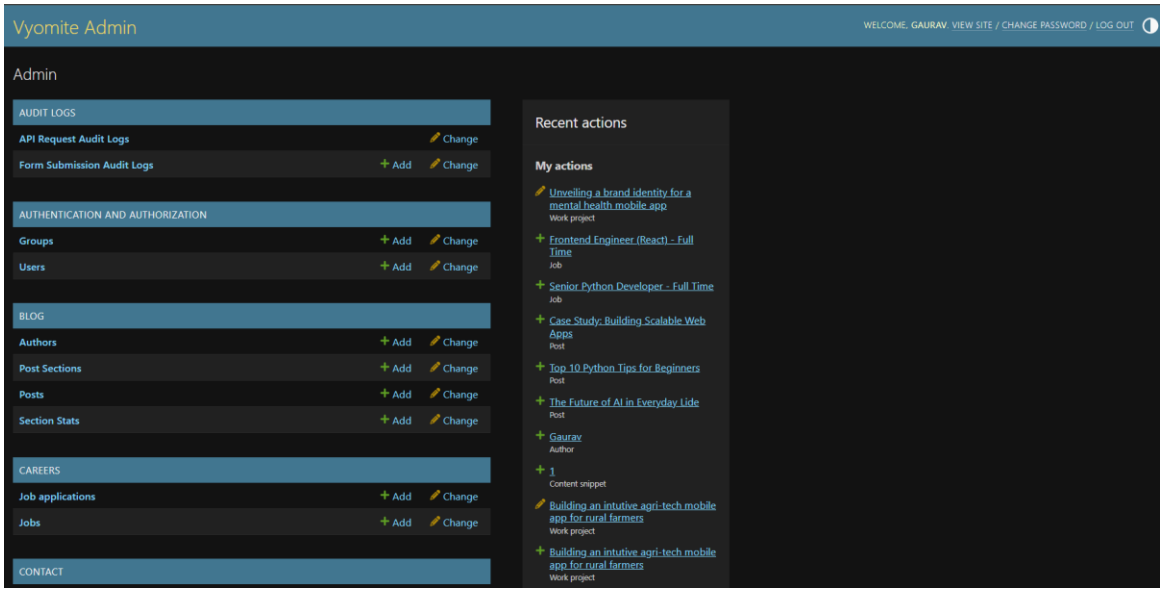
[4] DevCom, “Coding Standards and Best Practices Guide: Implementation Tips.” [Online]. Available: https://devcom.com/tech-blog/coding-standards-and-best-practices-guide-implementation-tips/.

APPENDIX

1. Task Management Platform Dashboard



2. Vyomite Website Swagger API Documentation



3. Vyomite Website Admin Dashboard